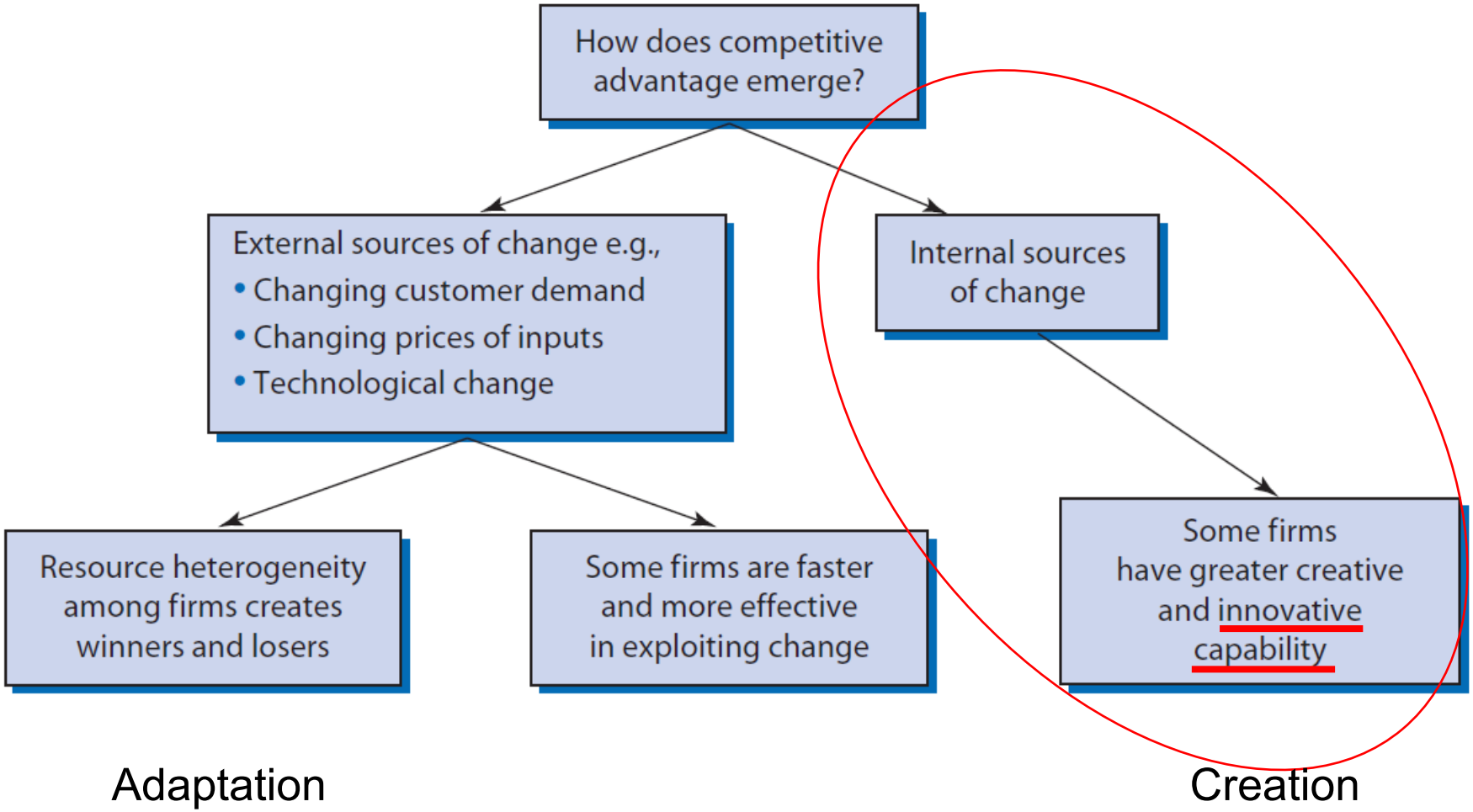
# STRATEGY IN TURBULENT ENVIRONMENTS

**Session 25**

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How does competitive advantage emerge?
External sources of change e.g.,
• Changing customer demand
• Changing prices of inputs
• Technological change
Internal sources of change
Resource heterogeneity among firms creates winners and losers
Some firms are faster and more effective in exploiting change
Some firms have greater creative and innovative capability
Adaptation
Creation

**1. Strategy in technology-intensive industry**

- 1.1. Market shaping vs. Adapting
- 1.2. The innovation process
- 1.3. Appropriation of value

**2. Strategies for navigating the riskiness of innovation**

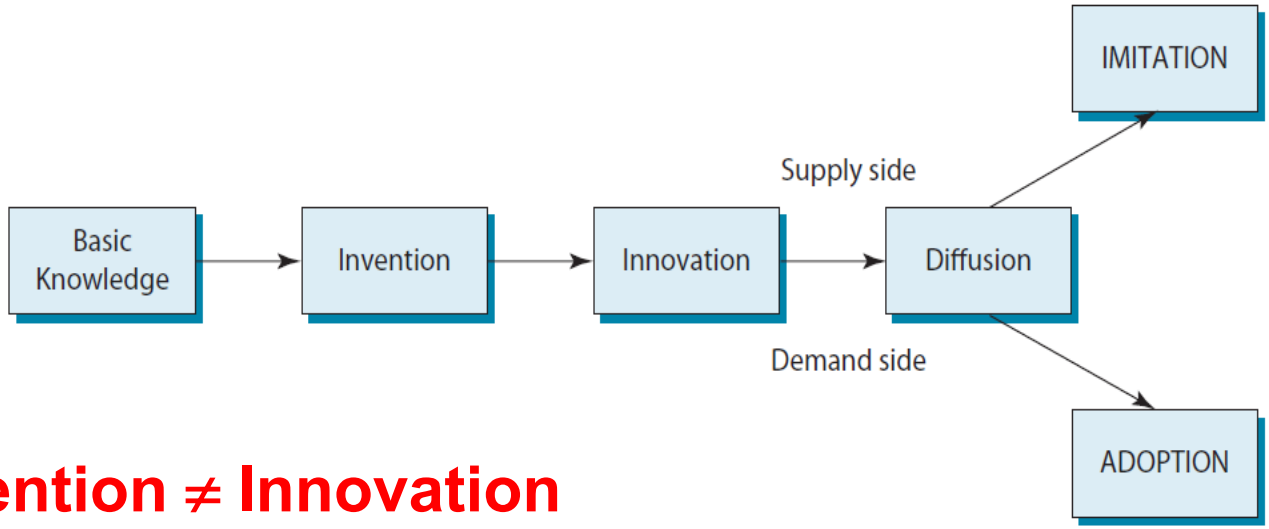
- 2.1. Incremental adaptive experimentation
- 2.2. Strategic adaptative experimentation
- 2.3. Big-bet resource investments (shaping)

**3. Strategy for technology-intensive industry with network effects**

- 3.1. Network effects
- 3.2. Tipping/coexistence
- 3.3. Platforms
- 3.4. Chicken & Egg
- 3.5. Complementors

| | Adaptation | Market shaping |
|---|---|---|
| **Strategic Logic** | **Respond to external changes** and uncertainties | **Proactively alter the market** structure or "rules of the game" |
| **View of the environment** | **Environment is given** and firms must fit | **Environment is malleable** and can be influenced |
| **Goal under uncertainty** | **Achieve flexibility** and resilience | Create **new sources of opportunity** or stability |
| **Typical actions** | **Experiment**, pivot, diversify, hedge, sense-and-respond | Set **new standards**, build ecosystems, influence regulation, educate customers |
| **Role of capabilities** | **Dynamic capabilities** for sensing and seizing opportunities | **Shaping capabilities** for orchestrating change and coordination |
| **Examples** | **Disney** adapting to streaming trends; **Toyota** adjusting supply chains | **Apple** shaping digital ecosystems; **Tesla** shaping the EV market and infrastructure |

*Source: Rindova & Courtney, 2020 AMR*




## Invention ≠ Innovation

- **Invention** = The creation of new products and processes through the development of **new knowledge** or from **new combinations** of existing knowledge
- **Innovation** = the initial **commercialization** of invention by producing and marketing a new good or service or by using a new method of production

# Innovations may involve no new technology



A pilot came up with the first "upright wheeled suitcase with two wheels on the bottom and a telescoping handle" in 1987

A huge success!



**Which firms made the most money in the computer industry?**



Microsoft

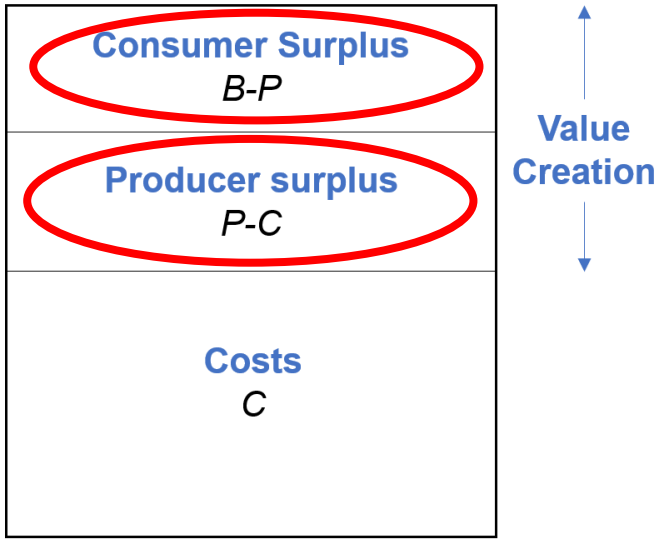
**Which firm invented most of the key foundations of modern computing?**

XEROX®

In the 1970s, Xerox for instance developed:

- A Graphical User Interface
- The mouse as standard input
- Ethernet networking

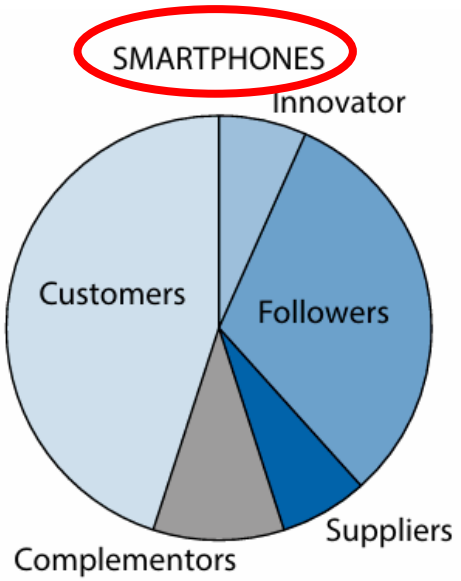
- Remember value creation (BPC framework) and value capture from Session 2?




- The innovator (focal firm, producer) does not necessarily capture all new value created from innovation

1. Value created by Innovation

2. Innovator's ability to appropriate a share of the value




The first smarphones were developed by IBM (1993) and Nokia (1996)…

The innovator (focal firm, producer) does not necessarily capture most of the value created

- Appropriability of value = Conditions that influence the distribution of the value created by innovation.
- **Strong regime** → innovator can capture a substantial share of the created value
- **Weak regime** → other parties capture a major share of the created value

**Components that help innovators to appropriate (capture) value**

1. Intellectual Property Rights
2. Complexity of the technology & Tacitness of knowledge.
3. Lead time. Innovate quickly
4. Complementary resources.



United States Patent [19]
Perini
[11] 4,327,877
[45] May 4, 1982

[54] WINDING DEVICE
[76] Inventor: Fabio Perini, Via della Fornace, S. Michele di Moriano, Lucca, Italy
[21] Appl. No.: 186,002
[22] Filed: Sep. 10, 1980
[30] Foreign Application Priority Data
Sep. 21, 1979 [IT] Italy
[51] Int. Cl.
[52] U.S. Cl.
[58] Field of Search
[56]

U.S. PATENT DOCUMENTS

| | | | |
|---|---|---|---|
| 2,676,764 | 4/1954 | Aslen | 242/66 X |
| 3,817,467 | 6/1974 | Dambroth | 242/66 |
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| 4,133,495 | 1/1979 | Dowd | |

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| | | | |
|---|---|---|---|
| 1435525 | 5/1976 | United Kingdom | 242/66 |

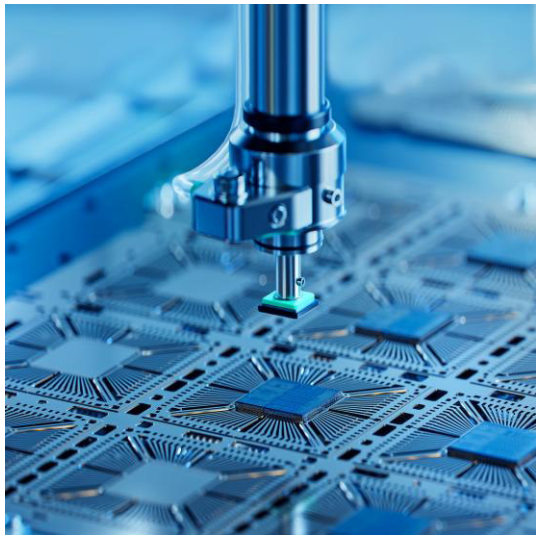
Primary Examiner—Edward J. McCarthy
Attorney, Agent, or Firm—McGlew and Tuttle

[57] ABSTRACT

A continuous winding device for webs of paper, such as ... comprises a first drum which is drivable at ... equal to the speed at which the mate... during the winding phase, and a ... the first by a distance corre... winding cores for the for... and drum is ... of the first, ... the sec... rying the ... replace... ed as is a ... between the two drums. ... is also used which bears ... and thus holds the spool ... second drums and the presser ... The device includes, the first ... with perforated zones, for receiving ... adapted to exert a suction force during ... removal of the finished roll and insertion of a new core. The material is thus brought to the zone in which the tearing is to take place, and the material is securely inserted between the drum and the new core.

12 Claims, 11 Drawing Figures

PATENTED



Coca-Cola

Copying advanced semiconductor manufacturing from TSMC is hard even without patents…

…it's complex and requires a lot of tacit process knowledge

Coca-Cola's recipe is codifiable, and can easily be copied

….if not protected by a trade secret

…and a trademark

# First mover vs. Early mover

If being a **FIRST MOVER is costly**, then it makes sense only if isolation mechanisms **can protect innovation** and lead to increased value capture.

**EARLY MOVER: "active waiting" to learn from others.**

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**2. Strategies for navigating the riskiness of innovation**

1. Incremental adaptive experimentation

2. Strategic adaptative experimentation

3. Big-bet resource investments (shaping)

**3. Strategy for technology-intensive industry with network effects**

# How to navigate innovation if it is risky?

## 1. Incremental adaptive experimentation

Booking.com

## 2. Strategic adaptative experimentation

Google



LOON™

## 3. Big-bet resource investments (shaping)

SPACEX

TESLA

*Source: Thomke, 2020 HBR*

# 1. Incremental adaptive experimentation

1. **Small experiments** (for example A/B testing) in large scale for adjustments in how the firm creates and captures value
2. Often **hypothesis about customer preferences**
3. Best firms run **multiple experiments at the same time**, consistently
4. Often involves a **culture of experimentation** and freedom to experiment at all levels of the firm

- For example, think about Booking.com:

Booking.com

*Source: Thomke, 2020 HBR*

## SCENARIO #1

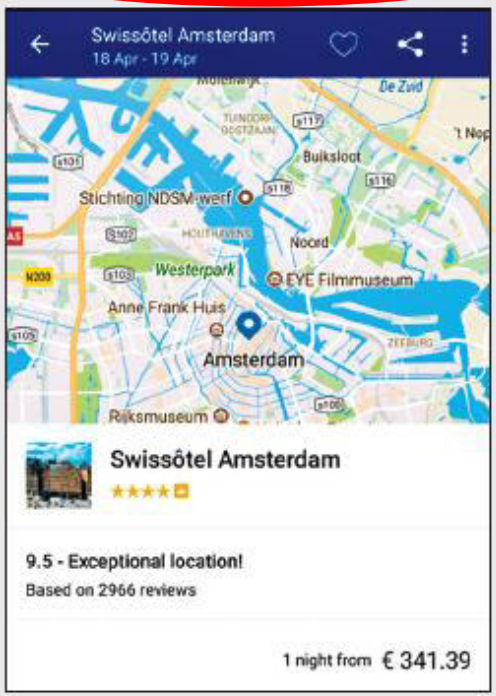
**Hypothesis**

Highlighting a neighborhood's walkability helps users make better decisions about property location.

### A

**The Control**

Shows the site's current practice




### B

**The Treatment**

Adds walkability information




**The Result**

The treatment had no significant impact on the key metric. The current practice is kept in place.

[Source: Thomke, 2020 HBR]

## SCENARIO #2

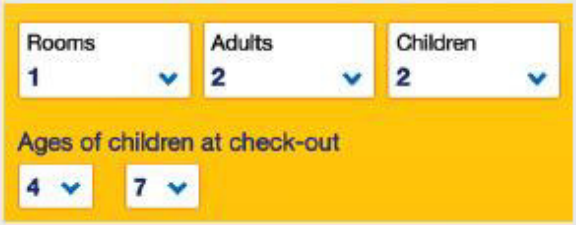
**Hypothesis**

Displaying the checkout date when users select the age of children in their party improves their experience.

### A

**The Control**

Shows the site's current practice




### B

**The Treatment**

Adds the checkout date above children's ages




**The Result**

The treatment had a significant positive impact on the key metric, and the change is implemented.

# 2. Strategic adaptative experimentation



**Real options approach**

- Invest in technologies/businesses just enough to "reserve the right" to continue development after uncertainty has decreased
- …or pivot …or terminate project

## Example: Smart glasses

Link to Google Glass

- Google developed smart glass tech, targeting consumers
- Consumers didn't want the product (in large scale) → pivot to businesses
- Manufacturing employees could work hands-free while being connected
- Not enough demand → Project discontinued (tech flows into VR/AR)
- Meta introduces smart glasses → Google reactivates its tech

# Google X has many of these…

One famous example is Loon, the stratospheric Internet balloons that was discontinued due to a challenging business case despite early successes in select regions like Kenya and Brazil




Link to Loon

# 3. Big-bet resource investments (shaping)

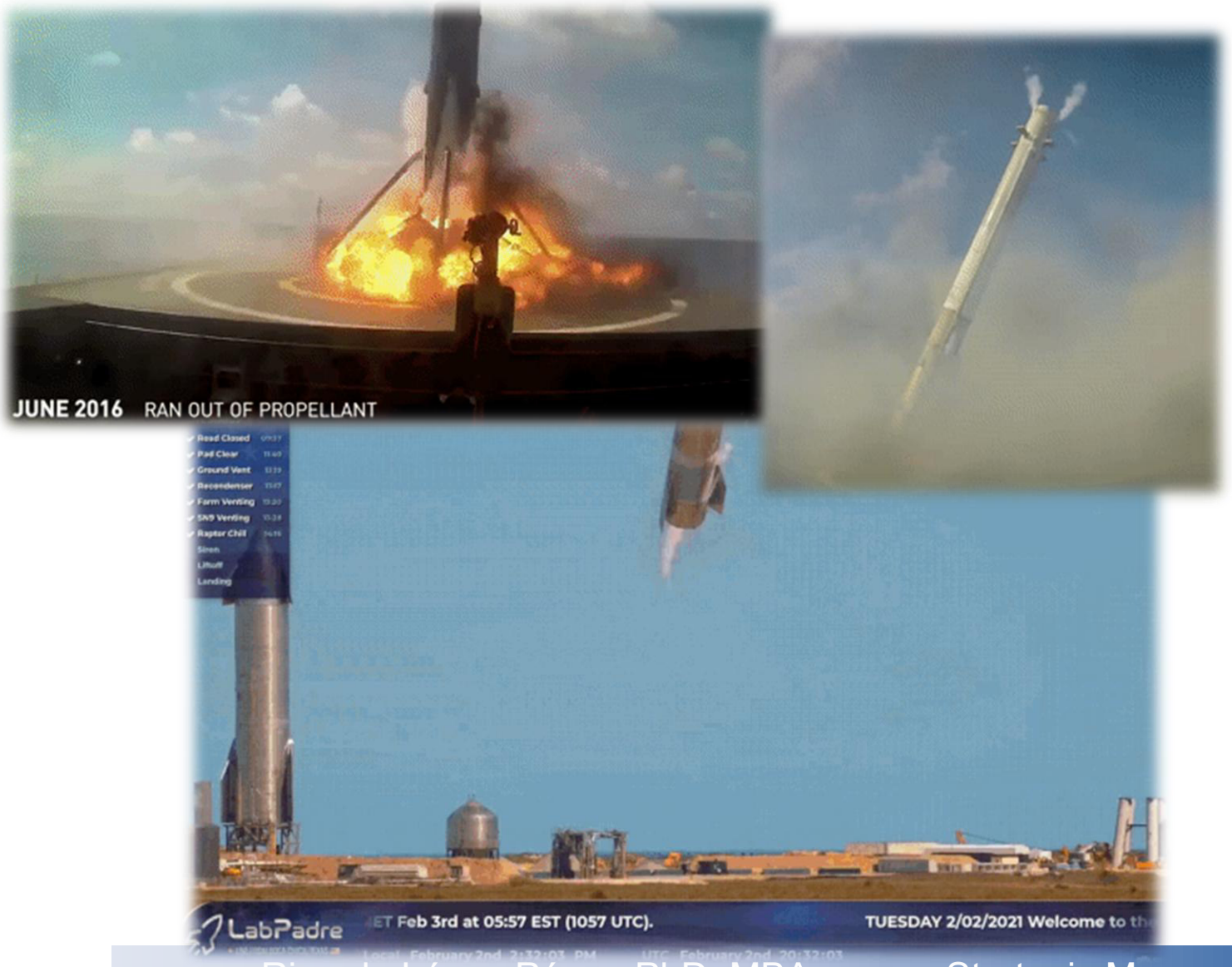
- Large investment in a tech or business to change current market structure or prevailing norms and assumptions
- Such a move persuades and signals commitment to partners, suppliers, customers
- Small-scale pivots still possible, but large investments are often irreversible due to the infrastructure that needs to be developed before innovations can even take place
- Using or creating uncertainty for one's own advantage
- Famous examples:
  - **Apple** iPhone+Ecosystem (iTunes, Appstore)
  - **Tesla** EVs+charging network + battery Gigafactories
  - **Microsoft Azure** for enterprise software from on-premise licensing to clou-based infrastructure + subscription model

SPACEX

- From discarded to reusable rockets → dramatic cost reduction
- Billion-dollar investment in vertical landing and reuse tech
- Experimentation now addressed ONLY as "How can we make it work?" NOT "What business or tech should we invest in?"
- Learning for shaping may require sacrifices: potential public humiliation, acceptance of risks and failures, need to prove yourself



SPACEX **failures**



SPACEX

**successful rocket landing**



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## 3. Strategy for technology-intensive industry with network effects

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# 1. Network Effect

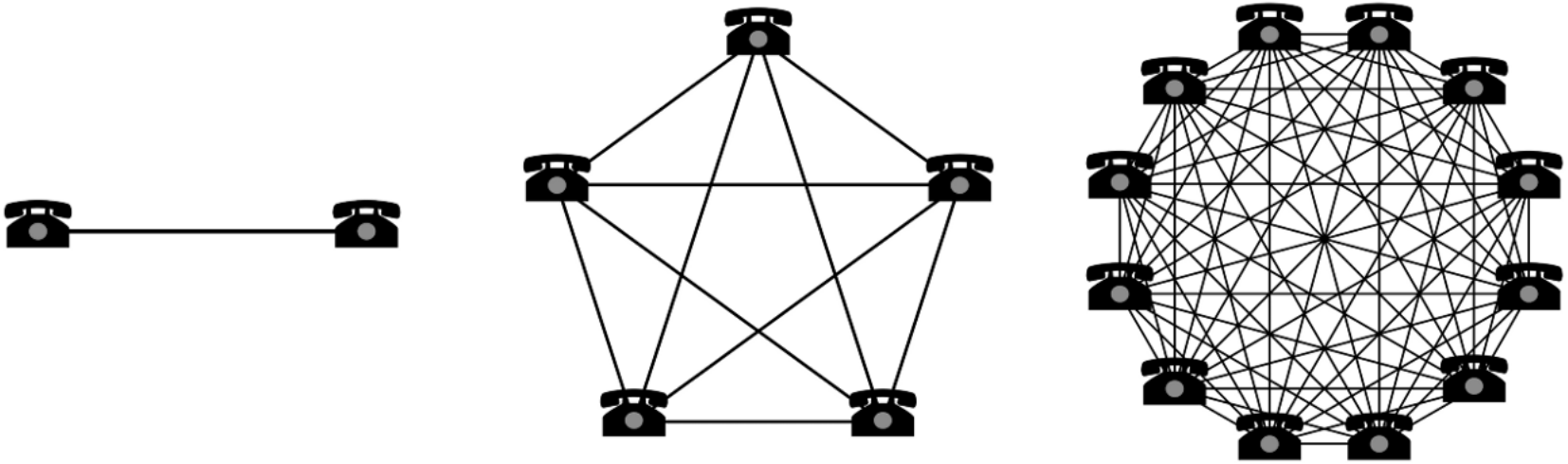
**Most of the world's most valuable firms run platform models with network effects**



# Direct network effects

Direct (same-side):

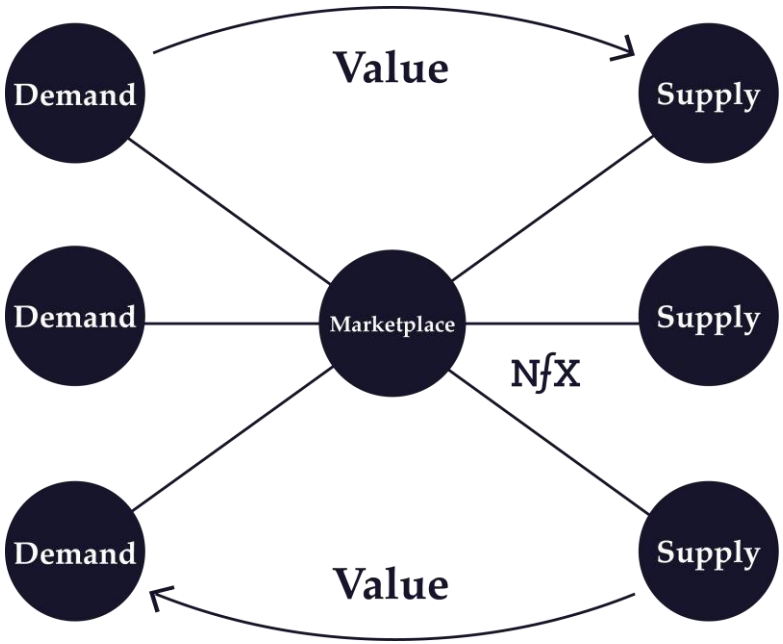
more users → more value per user (telephone, social network).

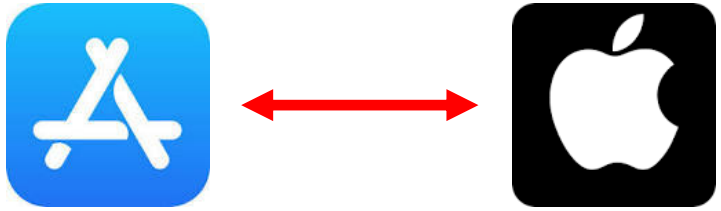


# Indirect network effects

Indirect (cross-side):

more of Side A raises value for Side B (consoles ↔ games; iOS ↔ apps).



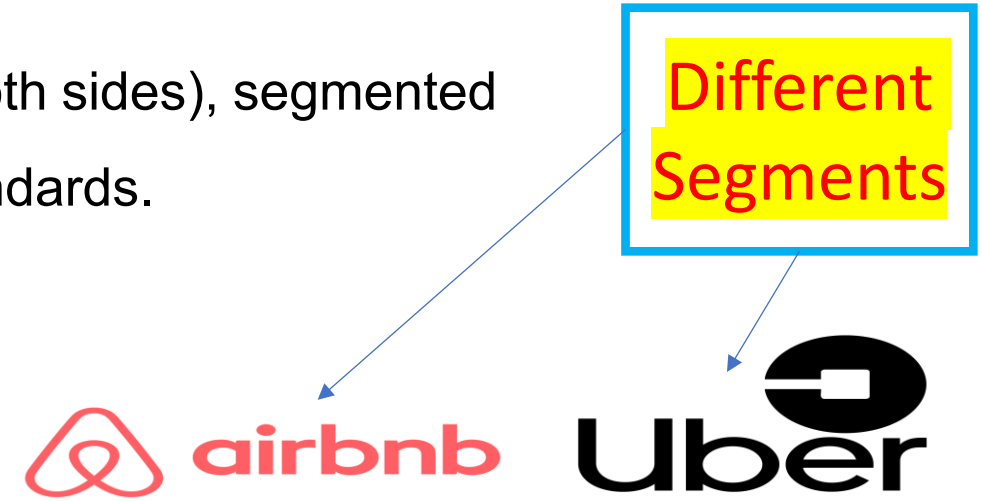


# 2. Tipping vs. Coexistence

- **Tipping / winner-take-most more likely:**
  - strong network effects + high multihoming/switching cost + low standalone differentiation.
  - **Global network structure**

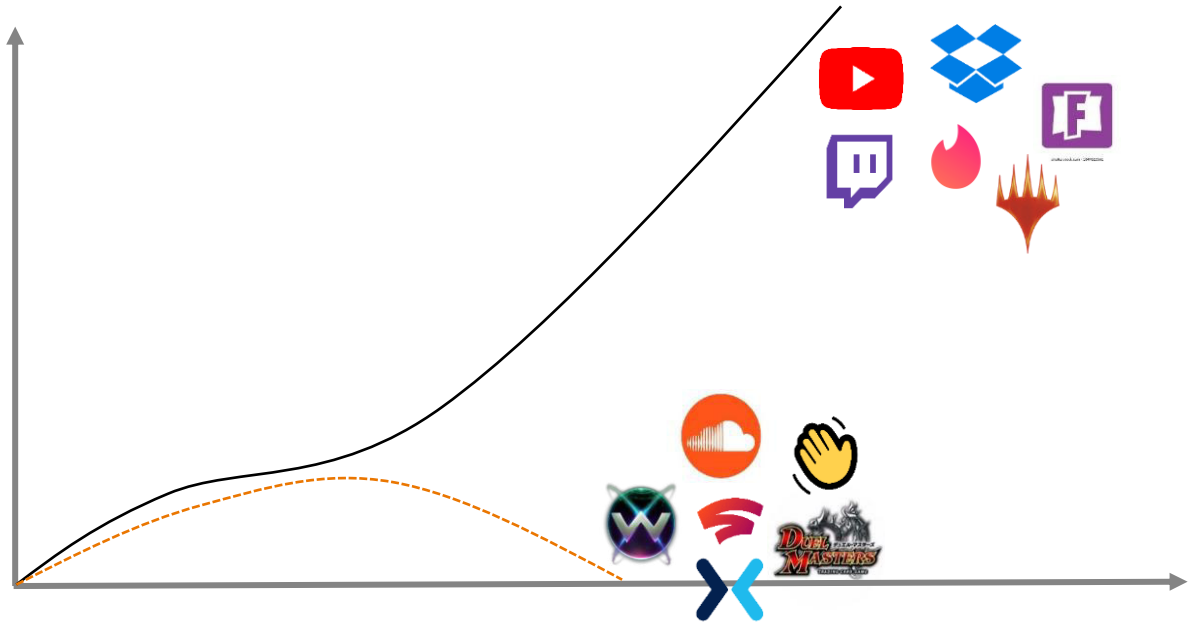



- **Coexistence more likely:**
  - cheap multihoming (one/both sides), segmented needs, or compatibility standards.
  - **Local network structure**

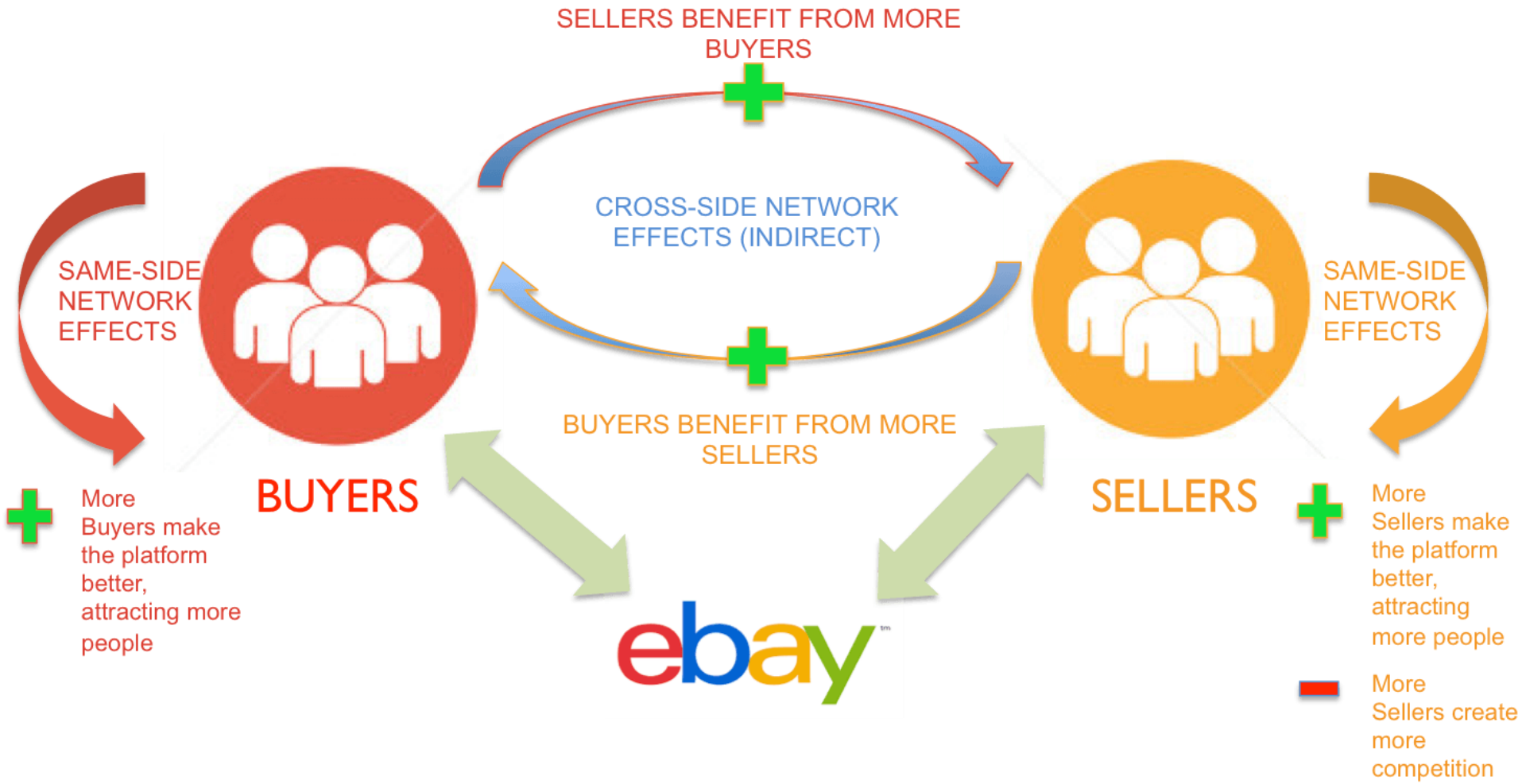


# 2. Tipping and winner-takes-all

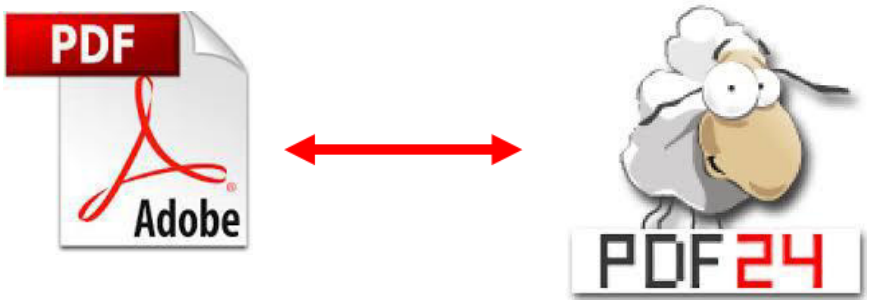
- Feedback: adoption → value → adoption; strength varies with design, quality, and governance.
- Network effects ≠ create an automatic monopoly, but tipping is more likely.



# 3. Platforms


SELLERS BENEFIT FROM MORE BUYERS
CROSS-SIDE NETWORK EFFECTS (INDIRECT)
BUYERS BENEFIT FROM MORE SELLERS
SAME-SIDE NETWORK EFFECTS
BUYERS
+ More Buyers make the platform better, attracting more people
SELLERS
SAME-SIDE NETWORK EFFECTS
+ More Sellers make the platform better, attracting more people
- More Sellers create more competition
ebay

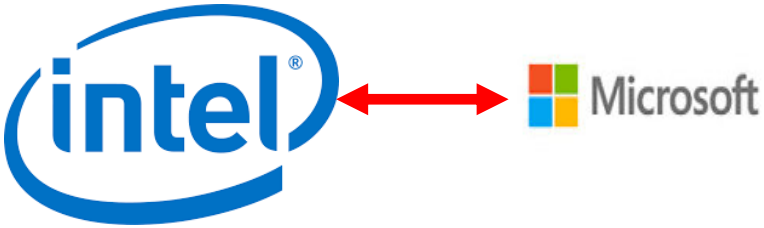
# 4. Chicken-and-egg problem

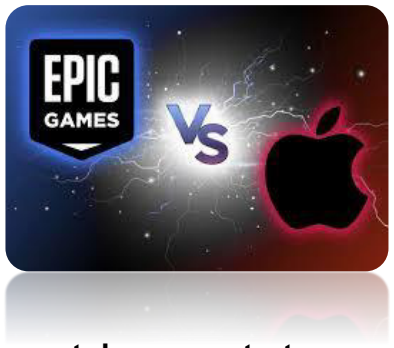



- Each side waits for the other—solve with price structure (not just price level).
- Subsidize the value-creating side (zero or negative price is sometimes optimal).
- Potential patterns: free consumers + paid complementors (ad-supported media); free SDKs/dev tools + paid users (enterprise).
- Non-price seeding: first-party, exclusive, or hit complements

Examples: free players vs. paid servers (Real/Adobe); console bundles; app store fee holidays.

# 5. Complementors: power, multihoming, ıce




- Complementors are co-strategists; power rises with must-have status and user access.
- Complementor multihoming weakens tipping; exclusivity can counteract—at a cost to variety.
- Governance levers: access rules, APIs, quality control; heavy-handedness invites backlash and regulation.
- Case: Apple v. Epic illustrates holdup risk and the control–goodwill trade-off.

**Thanks a lot!**

**Any questions?**